BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 057 · Iteration: FOR Loops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does selection do?

Q2. What does the Program Counter hold?
A. The next instruction's address
B. The last calculation result
C. All saved user files
D. Data moving from memory

Q3. “Larger cache increases the number of installed CPU cores.” Correct this. Explain why.

Q4. CPU A: 4 GHz, two cores, 2 MB cache. CPU B: 3 GHz, eight cores, 8 MB cache. Why is neither fastest for every task?

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I Do · Read new knowledge

Key vocabulary

FOR loop — Repeats a block for each count in a specified inclusive range.

Counter — The loop variable changes automatically each iteration.

NEXT — NEXT marks the end of a FOR loop and triggers the next iteration.

Counted iteration — Useful when the number of repetitions is known in advance.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

FOR loop: Repeats a block for each count in a specified inclusive range. For example, consider a daily exercise counter.

Counter: The loop variable changes automatically each iteration. For example, consider a quiz warm-up.

NEXT: NEXT marks the end of a FOR loop and triggers the next iteration. For example, consider a ticket numbering.

Counted iteration: Useful when the number of repetitions is known in advance. For example, consider a row display.

Worked example

A daily exercise counter is being developed. Explain how for loop and counter are used. [4 marks]

1. Identify where this particular program needs control or values.

2. FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].

3. Counter [1]; The loop variable changes automatically each iteration. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A quiz warm-up is being programmed. Explain why for loop matters. [2 marks]

Worked method: FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].

Guided 2

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

For a ticket numbering, explain how counter is used. [2 marks]

Worked method: Counter [1]; The loop variable changes automatically each iteration. [1].

Guided 3

A row display needs next and counted iteration. Explain the roles of both. [4 marks]

Worked method: NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1]. Counted iteration [1]; Useful when the number of repetitions is known in advance. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Fill the loop end value so "Back up" prints three times.

```
for attempt = 1 to ...
    print("Back up")
next attempt
```

write

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how counter and next can help a competition scoreboard. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of for loop, next and counted iteration when creating a maths drill. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Repeats a block for each count in a specified inclusive range.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. It chooses a branch using a condition. Selection can be nested.

Q2. A — The next instruction's address

Q3. A larger cache holds more often-used data and instructions. This can reduce slower memory access.

Q4. A has more cycles per second, which may help one task. B has more cores for parallel work and more cache to reduce slow memory access. Speed also depends on the task and CPU design.

I Do / We Do pages · Guided practice

Q1. FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1].

Q2. Counter [1]; The loop variable changes automatically each iteration. [1].

Q3. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1]. Counted iteration [1]; Useful when the number of repetitions is known in advance. [1].

You Do pages · Code First

```
for attempt = 1 to 3
    print("Back up")
next attempt
```

Expected result: Back up Back up Back up

You Do pages · Independent questions

Q1. FOR loop [1]; NEXT [1].

Q2. Counter [1]; The loop variable changes automatically each iteration. [1]. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1].

Q3. FOR loop [1]; Repeats a block for each count in a specified inclusive range. [1]. NEXT [1]; NEXT marks the end of a FOR loop and triggers the next iteration. [1]. Counted iteration [1]; Useful when the number of repetitions is known in advance. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: